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Clinical and echocardiographic study of hypertrophic cardiomyopathy in Tanzania

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TROPICAL DOCTOR 2006; **36:** 225–227

SUMMARY There are few published reports of hypertrophic cardiomyopathy (HCM) in Africans, partly due to lack of Echocardiography machines at most hospitals. Among 6680 patients referred for echocardiography at Muhimbili National Hospital between June 1998 and October 2002, 134 (0.19%) patients had HCM. Their mean age was 54.8 ± 14.2 years. In total 67.9% were men and 32.1% were women. Due to the diverse clinical features only eight (5.9%) patients had a correct diagnosis of HCM prior to their ECHO. The important role of echocardiography in the diagnosis of HCM is stressed with a plea for the increasing availability of this non-invasive technique for early and accurate diagnosis.

Introduction

Hypertrophic cardiomyopathy (HCM) is defined as the presence of a hypertrophied non-dilated left ventricle in the absence of cardiac or systemic diseases capable of producing significant hypertrophy.[1] Before the advent of echocardiography, clinical identification of HCM was virtually confined to those patients with a loud heart murmur (associated with the subaortic gradient). Unfortunately, this remains true to date in the developing countries of sub-Saharan Africa where echocardiography machines are not available in most hospitals. The objective of this study was to determine the demographic and echocardiographic features.

Methodology

Among 6680 patients referred for echocardiography at Muhimbili National Hospital in Dar es Salaam from district and regional hospitals between June 1998 and October 2002, 134 (0.19%) patients had HCM. These patients who were of African origin were consecutively and prospectively studied.

All patients had their history taken by experienced cardiologists. Physical examination included blood pressure, which was measured twice on the right arm using a mercury sphygmomanometer. Twelve lead electrocadiogram (ECG) and baseline laboratory studies were performed, including M-Mode and two-dimensional echocardiography as well as Doppler and colour flow mapping.

Echocardiographic evaluation was performed with a Hewlett-Packard Sonos 1000 scanner with a 2.5 MHz transducer. Patients were imaged while in a left lateral position in the echocardiography laboratory.

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Table 1 Echocardiographic findings in 134 patients with HCM and 72 control subjects

Parameter	*HCM (n=134)*	*Control subjects*	*P value*
• Age	54.8 ± 14.2	48.0 ± 11.1	NS
• LV end-diastolic diameter (cm)	4.7 ± 0.6	4.5 ± 0.4	NS
LV end-systolic diameter (cm)	2.8 ± 0.5	2.6 ± 0.3	NS
Interventricular septum thickness (cm)	2.1 ± 0.4	1.0 ± 0.1	0.001
LV posterior wall thickness (cm)	1.4 ± 0.6	0.9 ± 0.2	NS
Septum/posterior wall ratio	1.9 ± 0.5	1.0 ± 0.1	0.001
Left atrial diameter (cm)	4.4 ± 0.6	3.5 ± 0.2	0.05
LV ejection fraction (%)	72 ± 9	67 ± 10	NS
Fractional shortening (%)	42 ± 8	38 ± 6	NS
• Doppler diastolic indexes			
Peak E wave velocity (cm/sec)	40 ± 7	74 ± 4	0.05
Peak A wave velocity (cm/sec)	71 ± 7	36 ± 6	0.001
E/A wave ration	0.6 ± 0.5	2.4 ± 0.2	0.0001
LV outflow tract obstruction at rest (>30 mmHG)	38 (28.4%)		
LV outflow tract gradient (mmHg)	48 ± 32		
V-Max (M/s)	2.9 ± 1.5		
Systoic anterior motion of mitral valve	41 (30.6%)		
• Location of maximal hypertrophy			
Septum	68 (50.7%)		
Concentric	47 (35.1%)		
LV posterior wall	15 (11.2%)		
Apical hypertrophy	4 (3.0%)		

Data presented are mean value ± SD or number (%) of patients with hypertrophic cardiomyopathy (HCM)
LV=left ventricular
V-Max=maximum flow velocity in the LV outflow tract

Measurements were done according to the recommendations of the American Society of Echocardiography.[2]

Seventy-two age- and sex-matched asymptomatic individuals who came for medical examination for the purposes of travelling abroad, insurance or joining learning institutions were chosen as controls. All had echocardiography done in the same way as the patients.

Statistical analysis

All data are expressed as mean ± standard deviation. Statistical differences between groups were determined by unpaired *t*-tests. A *P* value of less than 0.05 was considered significant.

Results

The 134 patients had an age range from 22 to 91 years (mean 54.8 ± 14.2 years). Ninety-one (67.9%) patients were male and 43 (32.1%) were women. A total of 119 (88.8%) patients were symptomatic at the first presentation. The duration of symptoms was 2.1 ± 2.0 years. The main symptoms were: dyspnoea 84 (62.7%), chest pain 75 (55.9%), palpitations 68 (50.7%), pre-syncope 29 (21.6%) and syncope 12 (8.9%). The asymptomatic patients (11.2%) presented with incidental findings such as abnormal ECG ($n=6$), a new undiagnosed murmur ($n=5$) and abnormal echocardiography performed during evaluation for a major operation ($n=4$).

Due to diverse clinical features of HCM, the majority of the patients had wrong diagnoses on their referral notes to our echocardiography laboratory. Forty-two (31.3%) patients were diagnosed as having ischaemic heart disease, 40 (29.9%) as having dilated cardiomyopathy, 15 (11.2%) as having mitral valve prolapse and nine (6.7%) as having arrhythmias. Only eight (6.7%) patients had a correct diagnosis of HCM.

The pattern of hypertrophy among the 134 patients were: 68 (50.7%) had asymmetrical septal hypertrophy (ratio 1.9 ± 0.5), 47 (35.1%) patients had concentric hypertrophy, 15 (11.2%) had hypertrophy of the LV posterior wall and four (3.0%) had apical hypertrophy. The LP septal thickness, posterior wall thickness and left atrial diameter were significantly greater in patients with HCM than in control subjects ($P=0.001$). The LV end-diastolic diameter and LV end-systolic diameter did not differ significantly from the control group. However, the LV diastolic function was significantly impaired among the HCM patients than the control group ($P=0.0001$) (Table 1).

Discussion

HCM is wrongly thought to be a rare disease by the medical practitioners in developing countries. In an unselected hospital-based population evaluated for heart disease, the prevalence of HCM was 1:500 (0.17%).[3] In our study, the prevalence was 0.19%. Due to the diverse clinical features, three-quarters of the patients faced a delay between initial presentation to a primary care provider and the final diagnosis. Chest pain, dyspnoea and T-wave changes in the ECG were misinterpreted as coronary artery disease. Only eight (5.9%) patients had a correct diagnosis of HCM in their echo request forms. The correct diagnosis of HCM is of importance, otherwise some patients are admitted to the intensive care unit for suspected myocardial infarction.

Our patients were of relatively advanced age (54.8 ± 14.2 years) with only nine (6.3%) less than 30 years. Among patients with symptoms, 51 (38%) were asymptomatic until they were 60 or more years. The duration of symptoms among these patients was 2.1 ± 2.0 years. These observations underlie the importance of recognizing HCM as a disease compatible with advanced age and normal longevity. Our oldest patient was 91 years old. Such a perception is unappreciated in the available HCM published data.[1,4]

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The belief among medical practitioners in Sub-Saharan Africa that the obstructive form represents virtually the entire clinical spectrum of HCM should change. We now know that most HCM patients have the non-obstructive form. In our study, only 38 (28.4%) patients had LV outflow obstruction. With respect to the pattern of LV hypertrophy, asymmetrical septal hypertrophy was present in 50.7% of the patients, concentric LV hypertrophy in 31.5% and LV posterior wall in 11.2%. The typical apical HCM first described by Yamaguchi *et al.*[5] also occurs among Africans, and in our study four (3.0%) had apical HCM.

In conclusion, echocardiography is the method of choice for the diagnosis of HCM. A plea is therefore made for the increasing availability and use of this non-invasive technique for early and accurate diagnosis of HCM in sub-Saharan Africa. Africans are reluctant to allow autopsy and thus echocardiography not only will enable early diagnosis and therefore early treatment, but it will also allow a true assessment of the incidence of this disease in our area.

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Blood transfusion in an emergency paediatric unit in Owerri, Nigeria: pattern and outcome

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SUMMARY The increasing number of patients presenting with severe anaemia and who invariably receive blood transfusion is of concern. This retrospective study reports the pattern of presentation and the outcome of management of patients who were transfused in the Emergency Paediatric Unit of Federal Medical Centre, Owerri, between September 2002 and August 2004.

Introduction

Anaemia is a major cause of childhood morbidity and mortality in sub-Saharan Africa and it is usually multifactorial in origin. Severe anaemia (defined as haemoglobin less than 5 g/dL) is a common complication of childhood malaria; it is also seen in association with protein and micronutrient deficiencies notably iron and folic acid, helminthic infections and sickle cell disease.[1–4]

These patients with severe anaemia are usually admitted as emergencies and blood transfusion is invariably given as a life-saving measure in spite of the well-documented risks.[5]

This retrospective study was carried out to determine (a) the pattern of presentation of these patients and (b) the outcome of management.

Materials and methods

A review of the medical records of patients who were transfused in the Emergency Paediatric Unit of the Federal Medical Centre Owerri, Nigeria, between September 2002 and August 2004 was carried out. Information extracted included sex, age, blood group, pre-transfusion haemoglobin level and the outcome of management. Analysis of data was performed using SPSS version 10.

Result

During the reviewed period, 4574 patients were attended in the Emergency Paediatric Unit and 643 were treated with blood transfusion. This gives a transfusion rate of 14.1%. The medical records of 65 of these transfused patients were incomplete, so only 578 patients were analysed.

Age and sex

There were 337 (58.3%) males and 241 (41.7%) females with a male/female ratio of 1.4:1. In all, 481 (83.2%) of the transfused patients were under five years of age.

Blood group

In all, 557 (96.4%) were Rhesus (D) positive while 21 (3.3%) were Rhesus negative. Further over 334 (57.8%) were of blood group O, 164 (28.4%) of group A, 65 (11.2%) of group B and 15 (2.6%) were of group AB.

Pre transfusion haemoglobin level

Fifty-three (9.2%) had pre-transfusion haemoglobin less than 2.9 g/dL, while 341 (59%) had between 3 and 5 g/dL and 186 (31.8%) had levels greater than 5 g/dL.

Mortality

Sixty-six patients died giving a mortality rate of 11.4%. Mortality rate among under five years of age was 12.5% and 6.2% among those above five years of age. It was

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